

Juniorprogrammierer.de

Java 23: jdbc

2024/25/26 – Sascha Stojanovic

What are we doing today?

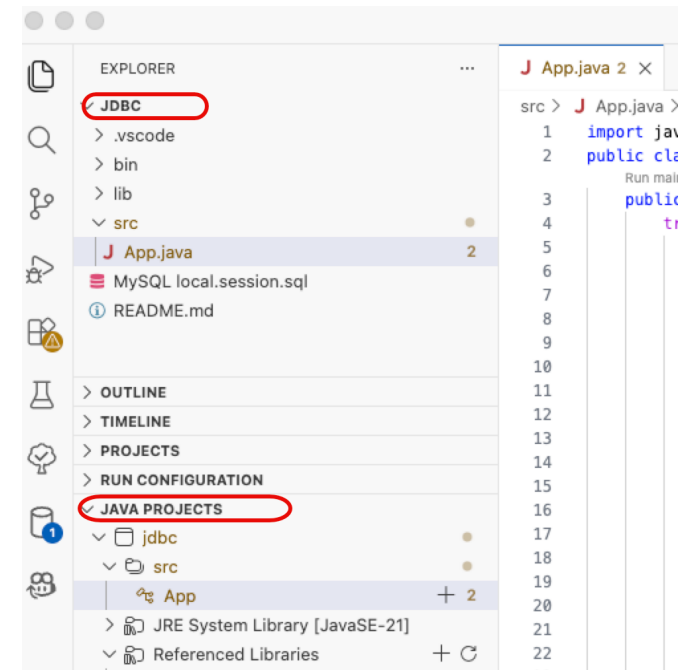
- Last lesson
 - We used Foreign Key to create a connection between two tables
 - We created a Select to those two tables
- Last step is to execute a Select via Java therefore we use JDBC

What is JDBC?

- JDBC (Java Database Connectivity) is an API (Application Programming Interface) for Java that allows Java applications to interact with various databases using SQL statements
- It consists of 3 main elements
 - Connection-Object: Represents a session with a specific database
 - Statement-Object: Is used to execute SQL queries against the database
 - ResultSet-Object: It holds the data retrieved from a database query. You can iterate through it to process the returned data

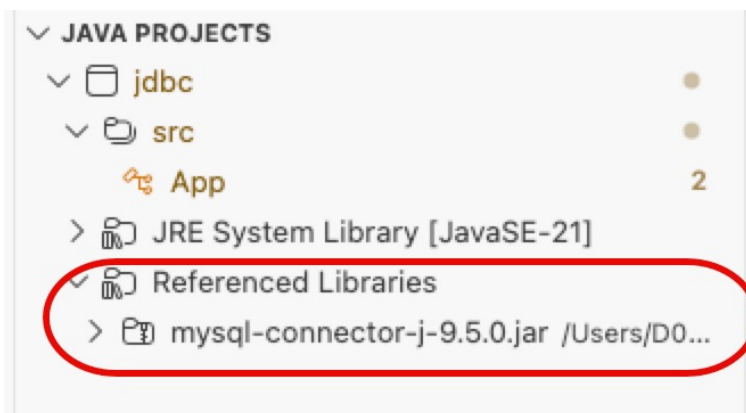
How to - I

- Download Jdbc:
<https://dev.mysql.com/downloads/connector/j/>
and unzip it (keep it in download-folder)
- Next in Vscode
 - Enter “>Java: Create Java Project ...” -> Choose “No build tools”
 - Choose a place to save your project (You can choose the recommendation)
 - Give your project a fitting name (e.g. “Jdbc”)
- Result:
 - A new Vscode window is opening of your project
 - And you got now a “Java Projects” register



How to - II

- We are almost done
- Create now a User-Relation consisting of “ID”, “firstname”, “lastname” and “email” and fill it with at least 4 values
- Now add code to your “App.java” file in “Jdbc” – project given by Sascha
- Last add in register “Java Projects” the mysql-connector-j...jar
- When running the “App.java” your should see a result in your terminal



```
D067975 .../Juniorprogrammierer_java/jdbc_example/jdbc 11:46 c
VirtualMachines/sapmachine-21.jdk/Contents/Home/bin/java @/var/folders/v9
Connection established.
Executing query: SELECT * FROM User
Lastname: Doe
Lastname: Smith
Lastname: Johnson
Lastname: Brown
```

Excercise

- Adapt the given Query of your “App.java” so you see all the content of your User-Table
- Adapt the given Query of your “App.java” so you insert a Tupel to your User-Table
- Adapt the given Query of your “App.java” so you delete a Tupel of your User-Table